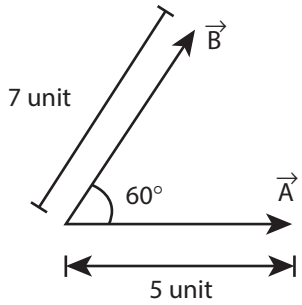
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EXAMPLE 2.1

Two vectors $\vec{A}$ and $\vec{B}$ of magnitude 5 units and 7 units respectively make an angle 60° with each other as shown below. Find the magnitude of the resultant vector and its direction with respect to the vector $\vec{A}$.

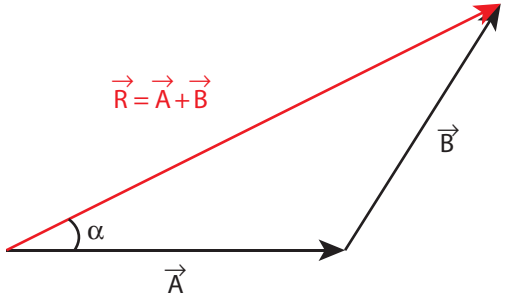


Solution

By following the law of triangular addition, the resultant vector is given by

$$\vec{R} = \vec{A} + \vec{B}$$

as illustrated below



The magnitude of the resultant vector $\vec{R}$ is given by

$$R = |\vec{R}| = \sqrt{5^2 + 7^2 + 2 \times 5 \times 7 \cos 60^\circ}$$

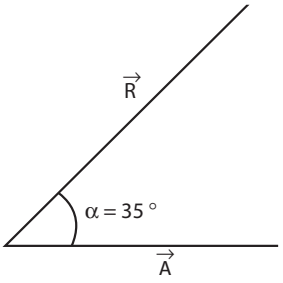
$$R = \sqrt{25 + 49 + \frac{70 \times 1}{2}} = \sqrt{109}\ units$$

The angle α between $\vec{R}$ and $\vec{A}$ is given by

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta} \qquad (2.2)$$

$$\tan\alpha = \frac{7 \times \sin 60}{5 + 7\cos 60} = \frac{7\sqrt{3}}{10 + 7} = \frac{7\sqrt{3}}{17} \cong 0.713$$

$$\therefore \alpha \cong 35^\circ$$



Note

Another method to determine the resultant and angle of resultant of two vectors is the Parallelogram Law of vector addition method. It is given in appendix 2.1

2.3.4 Subtraction of vectors

Since vectors have both magnitude and direction two vectors cannot be subtracted from each other by the method of ordinary algebra. Thus, this subtraction can be done either geometrically or analytically. We shall now discuss subtraction of two vectors geometrically using the Figure 2.19

For two non-zero vectors $\vec{A}$ and $\vec{B}$ which are inclined to each other at an angle θ, the difference $\vec{A} - \vec{B}$ is obtained as follows. First obtain $-\vec{B}$ as in Figure 2.19. The angle between $\vec{A}$ and $-\vec{B}$ is 180–θ.

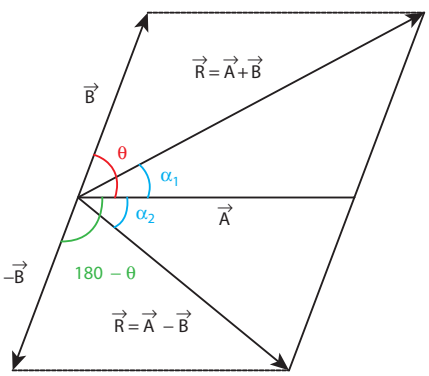


Figure 2.19 Subtraction of vectors